

EDUCATION

NIT Surat	B.Tech - Artificial Intelligence	CGPA: 7.4	2023–Present
SGVP, Rajkot	12th (CBSE)	87.1%	2023
Kendriya Vidyalaya, Rajkot	10th (CBSE)	91.2%	2021

RESEARCH

- Can You Break RLVER? Probing Adversarial Robustness of RL-Trained Empathetic Agents

Reinforcement Learning, LLM Evaluation, Empathy Modeling

– Designed adversarial personas and psychologically grounded dialogue trajectories for the **Adversarial Empathy Benchmark (AEB)**, including gaslighting, emotional escalation, and manipulation-based interactions.

– Conducted controlled evaluations across **480 adversarial dialogues** comparing RL-trained (PPO, GRPO) and baseline LLMs under identical conditions.

– Proposed the **Emotional Consistency Score (ECS)** to distinguish emotional state tracking from state improvement in empathetic dialogue systems.

– Observed improved emotional responsiveness, stronger hidden intention detection, and zero dialogue collapse in RL-trained agents under adversarial settings.

– **Manuscript under review at NeurIPS 2026.**

PROJECTS

- VidSynth AI – YouTube Chat Assistant

 [GitHub](#)

 [Live Demo](#)

– Independently developed the complete **VidSynth AI** system, including transcript processing, RAG pipeline integration, chatbot workflow, and deployment pipeline.

– Developed a **Streamlit-based RAG application** using **Gemini 2.5 Flash Lite** and **ChromaDB** for context-aware YouTube video question answering.

– Engineered transcript extraction, English translation, topic detection, and note generation modules using **LangChain** and **Sentence Transformers**.

– Designed and deployed a real-time interactive web application on **Hugging Face** for video summarization and conversational querying.

• Rare Gen Master - Synthetic Data Generation for Rare Cancer Diseases Using GANs

 [GitHub](#)

– Implemented a **GAN-VAE framework** on the Brain Glioma dataset to synthesize rare cancer imaging data, addressing dataset scarcity and class imbalance.

– Validated generated samples using **FID/IS metrics** and evaluated downstream classification performance improvements.

– Focused on generating privacy-preserving synthetic medical data for safe healthcare ML experimentation.

– **Research paper accepted at IEEE AICAS 2026** for this work on GAN-VAE based synthetic rare cancer data generation for rare brain cancer imaging datasets.
- SKILLS
- Programming Languages: C, C++, Python

APIs: REST APIs , FastAPI

Databases: MongoDB, MySQL , PostgreSQL

AI Domains: Machine Learning, Deep Learning, Reinforcement Learning, NLP, RAG Systems, GANs, VAEs

AI/ML Libraries & Frameworks: PyTorch, Scikit-learn, TensorFlow, Hugging Face Transformers, LangChain, ChromaDB

Data Science & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Tools & Platforms: Git, Colab, Kaggle, GitHub, Jupyter Notebook, Anaconda, VS Code
- POSITIONS OF RESPONSIBILITY
- Manager, MINDBEND (Annual Technical Festival)

Jan 2026 – Apr 2026

• Co-Head, MINDBEND (Annual Technical Festival)

Jan 2025 – Apr 2025

• Executive, MINDBEND (Annual Technical Festival)

Jan 2024 – Apr 2024